ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

FINAL MODEL: SENTIMENT ANALYSIS USING TENSORFLOW BIDIRECTIONAL LSTM

1. Introduction

The final model is developed using **deep learning techniques** to improve sentiment classification performance over traditional machine learning models. This model leverages a **Bidirectional Long Short-Term Memory (BiLSTM)** network implemented using TensorFlow/Keras to analyze student feedback more effectively.

Unlike earlier models, this approach captures the **context and sequence of words**, leading to more accurate predictions.

2. Model Overview

The model follows a deep learning pipeline consisting of:

- Text vectorization using TensorFlow
- Word embedding representation
- Bidirectional LSTM layers for sequence learning
- Dense layers for classification

The model classifies feedback into:

- Positive
- Negative
- Neutral

3. Feature Representation

TextVectorization

- Converts raw text into integer sequences
- Maintains word order
- Limits vocabulary size for efficiency

Embedding Layer

- Maps each word into a dense vector space
- Captures semantic relationships between words
- Enables the model to understand similarity between terms

4. Model Architecture

The model is built using the following layers:

Embedding Layer

Transforms tokens into dense vectors for learning word relationships.

Spatial Dropout

Reduces overfitting by randomly dropping entire feature maps.

Bidirectional LSTM Layers

- Processes text in both forward and backward directions
- Captures full contextual meaning of sentences
- Helps understand complex language patterns

Dense Layer

Extracts higher-level features from learned representations.

Dropout Layer

Prevents overfitting by randomly deactivating neurons.

Output Layer (Softmax)

Produces probability distribution across three sentiment classes.

5. Working Principle

- Input text is converted into sequences
- Embedding layer learns word representations
- BiLSTM captures contextual dependencies
- Dense layers refine features
- Softmax layer outputs final sentiment prediction

This allows the model to understand:

- word relationships
- sentence structure
- contextual meaning

6. Training Process

The model is trained using:

- **Adam optimizer** for efficient learning
- **Categorical Crossentropy loss** for multi-class classification

Additional techniques:

- Early stopping to prevent overfitting

- Learning rate reduction for better convergence

7. Model Performance

The model achieved approximately **85% accuracy**, showing a significant improvement over baseline models.

Observations:

- Better handling of **neutral class**
- Improved understanding of complex sentences
- More balanced predictions across all classes

8. Advantages of the Model

- Captures **context and word order**
- Handles complex linguistic patterns
- Higher accuracy compared to traditional models
- Better generalization on unseen data

The final model demonstrates the effectiveness of deep learning techniques in sentiment analysis. By using a BiLSTM architecture, the model successfully captures contextual information and improves prediction accuracy. This model represents a significant advancement over baseline approaches and provides a robust solution for analyzing student feedback.

RESULT

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	**0.84**	**0.89**	**0.87**	**1779**
Neutral	**0.82**	**0.89**	**0.85**	**1752**
Positive	**0.88**	**0.76**	**0.82**	**1782**

Overall Performance

Metric	Value
Accuracy	0.85
Macro Avg F1	0.84
Weighted Avg F1	0.84

Confusion Matrix

Actual \ Predicted	Negative	Neutral	Positive
Negative	1587	107	85
Neutral	102	1554	96
Positive	200	231	1351